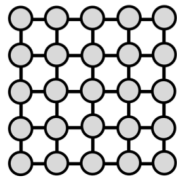


Geometric Deep Learning

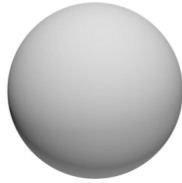
Final Project Guide

Academic Year 2025–2026

Last updated: March 11, 2026



Grids



Groups



Graphs



Geodesics & Gauges

This document contains everything you need to know about the final project component of the Geometric Deep Learning module. **Read it carefully** and refer back to it throughout the project. If anything is unclear, don't hesitate to ask and send me an e-mail at srivera@comillas.edu. As a first time read, it's important you dedicate 10-15 minutes to familiarize yourself with the content of this guide so you may understand the scope and what we are looking as part of this evaluation.

1. Overview

The final project accounts for **30%** of your total module grade, broken down as follows:

Component	Weight (of project)	Weight (of module)
Written report (PDF via $\text{\LaTeX}$)	67%	20%
Oral defense (1 slide)	33%	10%

The project is your opportunity to engage deeply with a topic in geometric deep learning. It can be primarily experimental, primarily theoretical, or a mixture of both. Regardless of the balance, the written report should rigorously document your work, your reasoning, and your findings.

2. Objectives and Expectations

This is a postgraduate-level project. The goal is not to reproduce a tutorial or re-implement a well-known baseline, it is to **investigate a research question**. Concretely, your project should:

1. **Pose a clear research question or hypothesis.** For example: “Does adding virtual nodes mitigate over-squashing in molecular property prediction tasks?” or “Can $\text{SE}(3)$ -equivariant convolutions improve pose estimation over standard CNNs on [dataset]?”
2. **Situate the question in context.** A brief but well-referenced literature review that explains why this question matters and what prior work exists.

3. **Propose and execute a methodology.** This may be an experimental pipeline, a theoretical analysis, or both.
4. **Present and discuss results honestly.** Negative results are acceptable and can earn high marks if analysed thoughtfully. Do not fabricate or cherry-pick, this will lower your grade.
5. **Reflect on limitations and future directions.**

Important

We expect you to invest real effort into this project. It should feel like a piece of work you are proud of, something you could show a future employer or include in a portfolio. That said, we recognize you are balancing multiple modules; the scope should be achievable within the allotted time. Deadline extensions may not be granted unless they can be properly justified.

3. Deliverables

You must submit the following by the deadline:

1. **Written report** (PDF, compiled from $\text{\LaTeX}$, **in English**). See Section 4 for format requirements.
2. **Source code**, submitted as a compressed archive (.zip) or a link to a Git repository. See Section 5 for requirements.
3. **One presentation slide** (PDF or PPTX) for the oral defense. See Section 6.

4. Report Format and Requirements

4.1 Template

Your report must be written in $\text{\LaTeX}$ using the **NeurIPS 2025 style template**, available at:

<https://media.neurips.cc/Conferences/NeurIPS2025/Styles.zip>

We recommend using **Overleaf** (<https://www.overleaf.com>) for real-time collaborative editing, though any $\text{\LaTeX}$ workflow is acceptable as long as you use the correct template and produce the expected output deliverable.

4.2 Page Limit

Section	Limit
Main body (introduction through conclusion)	6–8 pages
References	Unlimited
Appendix (optional: supplementary proofs, figures, tables)	Up to 4 pages

The page limit forces you to write concisely and prioritize what matters. Padding your report with filler, redundant figures, or excessive background **will be penalized**.

4.3 Expected Structure

While you have flexibility in organizing your report, a standard structure is recommended:

1. **Title, author(s), abstract** (~150–250 words).
2. **Introduction.** Motivate the problem and state your research question clearly.
3. **Related work.** Brief, focused literature review with proper citations.
4. **Background / Preliminaries.** Define notation and key concepts the reader needs.
5. **Method.** Describe your approach (architecture, theoretical framework, experimental setup).
6. **Experiments / Analysis.** Results, tables, figures, ablations, or proofs.
7. **Discussion.** Interpret your results, acknowledge limitations, suggest future work.
8. **Conclusion.** Summarize contributions in 1–2 paragraphs.
9. **References.**

For purely theoretical projects, sections 5–6 may be replaced with “Main Results” and “Proofs” as appropriate.

4.4 Quality Standards

- All claims should be supported by evidence (experimental or formal).
- Figures and tables must have captions, axis labels, and legends.
- Mathematical notation must be consistent and clearly defined.
- References must follow a consistent citation style (e.g., the default NeurIPS `natbib` style).
- The writing should be clear, precise, and free of grammatical errors.

5. Code Requirements

- Submit your code as a **GitHub/GitLab repository** (preferred) or a `.zip` archive.
- Include a **README** file explaining: what the code does, how to install dependencies, and how to reproduce your main results.
- Include a **requirements file** (`requirements.txt`, `environment.yml`, or equivalent).
- **Reproducibility matters.** If the graders cannot run your code, you will lose marks. Provide clear instructions, random seeds where applicable, and expected runtimes.
- You are free to use any framework (PyTorch, JAX, TensorFlow, etc.) and any relevant libraries (PyTorch Geometric, DGL, e3nn, etc.).

Purely theoretical projects. If your project involves no experiments, the code requirement is relaxed. You should instead include extended derivations, proofs, or formal analyses in the report appendix. If you wrote any supporting scripts (e.g., symbolic computation, visualization, verification), include them.

6. Oral Defense

6.1 Format

- You will prepare exactly **one slide** (PDF or PPTX) summarizing your project.
- You will have **5 minutes** to present, followed by **3–5 minutes of questions**.
- The one-slide constraint is deliberate: it forces you to distill your work to its essence.

6.2 What We Are Looking For

The presentation is important, but the Q&A is where we primarily assess depth of understanding. You should be able to:

- Explain *why* you made the design choices you made.
- Discuss limitations and what you would do differently.
- Answer technical questions about your method, your results, and the underlying theory.
- Demonstrate that you genuinely understand the work in the report.

Important

The oral defense is a critical integrity check. If you cannot explain the contents of your own report during the defense, this will be reflected significantly in your grade.

7. Group Policy

Projects may be completed **individually** or in **pairs** (groups of 2). If you work in a pair:

- You must include a **contribution statement** in the report (before references) clearly describing what each member did.
- During the oral defense, each member will be **questioned individually**. Your grade for the defense component is individual.
- Pairs are expected to produce work of **greater scope and depth** than individual projects.

8. Grading Rubric

Your report will be evaluated along four axes. The oral defense is graded separately.

8.1 Report (20% of module grade)

Criterion	Weight	Description
Technical depth & correctness	35%	Are the methods sound? Are experiments well-designed or proofs correct? Is the mathematical treatment rigorous?
Clarity & writing quality	25%	Is the report well-written, well-structured, and easy to follow? Is notation consistent? Are figures informative?
Experimental methodology or theoretical contribution	25%	For experimental work: proper baselines, ablations, statistical rigor, reproducibility. For theoretical work: novelty and significance of results.
Ambition & scope	15%	How challenging was the problem? How far did the student go beyond the baseline expectations? (See Section 8.3.)

8.2 Oral Defense (10% of module grade)

Criterion	Weight	Description
Presentation clarity	30%	Is the slide well-designed? Can the student communicate the core idea in 5 minutes?
Depth of understanding	50%	Can the student answer technical questions, explain design choices, and discuss limitations?
Critical reflection	20%	Does the student demonstrate honest assessment of their work's strengths, weaknesses, and potential extensions?

8.3 Ambition and Scope

The “ambition and scope” criterion rewards students who push beyond the safe and familiar. Ways to score highly here include (but are not limited to):

- Working on a topic **beyond standard GNNs** (e.g., group-equivariant CNNs, gauge-equivariant networks, geometric flows on manifolds, geodesic methods). The course has emphasized GNNs, so venturing further demonstrates breadth and initiative.
- Tackling a **theoretically challenging** question with formal analysis.
- Applying geometric deep learning to an **unconventional domain** with thoughtful justification.
- Producing a **novel contribution** (even a small one) rather than a pure reproduction.

This criterion is *not* a penalty for choosing a GNN topic. A well-executed, ambitious GNN project can score full marks on ambition. It is simply the case that going beyond GNNs is one natural way to demonstrate broader engagement with the field.

9. Suggested Project Topics

Below is a curated list of project ideas organized by theme. You may choose one of these, adapt one, or propose your own topic (subject to approval; see Section 10).

9.1 Graph Neural Networks

GNN-1: Expressivity and the Weisfeiler–Leman Hierarchy

Investigate the practical implications of GNN expressivity bounds. For example: construct graph pairs that are indistinguishable by 1-WL but distinguishable by higher-order methods, and evaluate whether higher-order GNNs (e.g., k -GNNs, PPGN) actually improve performance on tasks where this distinction matters.

Seed references: Xu et al., “How Powerful are Graph Neural Networks?” (ICLR 2019); Morris et al., “Weisfeiler and Leman Go Neural” (AAAI 2019).

GNN-2: Over-Squashing and Graph Rewiring

Study the over-squashing phenomenon in message-passing GNNs. Analyze how graph topology (e.g., bottlenecks, curvature) affects information flow, and evaluate rewiring strategies (e.g., via Ricci curvature, spectral gap optimization, or virtual nodes).

Seed references: Alon & Yahav, “On the Bottleneck of Graph Neural Networks” (ICLR 2021); Topping et al., “Understanding Over-Squashing and Bottlenecks on Graphs via Curvature” (ICLR 2022).

GNN-3: GNNs for Molecular Property Prediction

Apply GNNs to molecular graphs for property prediction (e.g., using QM9, ZINC, or OGB-MolPCBA). Compare message-passing approaches with more expressive architectures (e.g., transformers on graphs, subgraph GNNs). Focus on a specific molecular property and analyze what structural features the model learns.

Seed references: Gilmer et al., “Neural Message Passing for Quantum Chemistry” (ICML 2017); Hu et al., “Open Graph Benchmark” (NeurIPS 2020).

GNN-4: Heterogeneous or Temporal Graphs

Extend standard GNN architectures to handle heterogeneous graphs (multiple node/edge types) or temporal/dynamic graphs. Apply to a domain such as citation networks, knowledge graphs, or traffic prediction.

Seed references: Schlichtkrull et al., “Modeling Relational Data with Graph Convolutional Networks” (ESWC 2018); Rossi et al., “Temporal Graph Networks” (ICML 2020 Workshop).

9.2 Beyond Graph Neural Networks

GEO-1: Group-Equivariant Convolutional Networks

Implement and evaluate group-equivariant CNNs (e.g., $p4$, $p4m$ groups) on image classification or other structured data. Compare with standard CNNs in terms of sample efficiency and generalization, particularly under data augmentation regimes that simulate the relevant symmetries.

Seed references: Cohen & Welling, “Group Equivariant Convolutional Networks” (ICML 2016); Weiler & Cesa, “General $E(2)$ -Equivariant Steerable CNNs” (NeurIPS 2019).

GEO-2: $SE(3)$ -Equivariant Networks for 3D Data

Explore $SE(3)$ - or $E(3)$ -equivariant architectures (e.g., TFN, SEGNN, EGNN) for tasks on 3D point clouds or molecular conformations. Investigate where equivariance helps versus hurts compared to augmentation-based approaches.

Seed references: Thomas et al., “Tensor Field Networks” (arXiv 2018); Satorras et al., “ $E(n)$ Equivariant Graph Neural Networks” (ICML 2021).

GEO-3: Geometric Learning on Meshes and Manifolds

Work with data on meshes or manifolds: e.g., shape analysis, surface classification, or cortical brain data. Explore intrinsic approaches using geodesic distances, Laplacian eigenfunctions, or gauge-equivariant mesh CNNs.

Seed references: Bronstein et al., “Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges” (2021); de Haan et al., “Gauge Equivariant Mesh CNNs” (ICLR 2021).

GEO-4: Geometric Flows and Diffusion on Graphs

Investigate diffusion-based approaches on graphs (e.g., graph neural diffusion, GRAND, Beltrami flow). Analyse the connections between GNNs and PDEs on graphs, and study how the choice of diffusion operator affects downstream task performance.

Seed references: Chamberlain et al., “GRAND: Graph Neural Diffusion” (ICML 2021); Chamberlain et al., “Beltrami Flow and Neural Diffusion on Graphs” (NeurIPS 2021).

9.3 Theoretical / Analytical Projects

TH-1: Analyzing Spectral vs. Spatial GNNs

Provide a formal comparison of spectral and spatial graph convolution approaches. Prove equivalences or separations under specific conditions (e.g., graph regularity, filter order). Complement with small-scale experiments if appropriate.

Seed references: Defferrard et al., “Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering” (NeurIPS 2016); Balcilar et al., “Analyzing the Expressive Power of Graph Neural Networks in a Spectral Perspective” (ICLR 2021).

TH-2: Symmetry Breaking and Positional Encodings

Study the role of positional and structural encodings in GNNs. Analyze formally how different encodings (Laplacian eigenvectors, random walk probabilities, random features) affect expressivity. Under what conditions do they provably increase power beyond 1-WL?

Seed references: Dwivedi et al., “Benchmarking Graph Neural Networks” (JMLR 2023); Lim et al., “Sign and Basis Invariant Networks for Spectral Graph Representation Learning” (ICLR 2023).

10. Proposing Your Own Topic

If none of the suggested topics appeal to you, you are welcome to propose your own. To do so:

1. Write a short proposal (~200 words) describing: the topic, the research question, why it is relevant to geometric deep learning, and your planned approach.
2. Submit it as soon as possible via email to srivera@comillas.edu
3. I will review it and either approve, suggest modifications, or (rarely) redirect you to a different topic if the proposed one is out of scope, too easy, or infeasible.

Custom topics are encouraged and can be excellent.

11. Policy on AI Tools (LLMs, Code Assistants)

You are **permitted** to use AI assistants (e.g., ChatGPT, Claude, GitHub Copilot) as tools during your project. However:

1. **Disclosure is mandatory.** Include a short paragraph in your report (before references) describing which AI tools you used and how (e.g., “We used GitHub Copilot for code autocompletion and Claude for debugging error messages”).
2. **Understanding is mandatory.** The oral defense is where we verify that you genuinely understand your work. If you cannot explain code you submitted or concepts in your report, your grade will reflect this.
3. **AI-generated text must not be copied verbatim into the report.** You may use AI tools to help you draft, but the final text must be in your own words and reflect your own understanding.
4. **You are responsible for correctness.** If an AI tool generates incorrect code, wrong mathematics, or fabricated references, you bear full responsibility.

Important

Think of AI tools the way you would think of a calculator in a maths exam that allows one: it can speed up the mechanical parts, but you still need to know what to compute and why. The oral defense is designed to separate those who understand their work from those who do not.

12. Academic Integrity

Standard university policies on academic integrity apply in full. In addition:

- **Plagiarism** of text, code, or ideas without attribution will result in a mark of zero.
- **Fabrication of results** (inventing data, faking experiments) will be treated as a serious offense. Remember: the conclusion of your results is not correlated with grade, well-analyzed negative outcomes can end up with a better grade than poorly posed research questions.
- **Proper attribution** is expected. If you build on someone else's codebase, cite it. If you adapt an architecture from a paper, reference it.
- For group projects, both members are jointly responsible for the integrity of the entire submission, no matter who incurred in the offense.

13. Frequently Asked Questions

Can I work on a Kaggle competition as my project?

Only if you frame it as a research investigation (with a clear question, methodology, and analysis), not as a leaderboard chase. A report that simply says “we tried X architectures and got Y scores” will not score well.

Do I need to beat state-of-the-art results?

No. We are evaluating the quality of your investigation, not the magnitude of your numbers. Thoughtful negative results with good analysis can earn higher marks than marginal SOTA improvements with poor understanding.

Can I use a pre-existing codebase as a starting point?

Yes, as long as you clearly attribute it and your contribution goes meaningfully beyond it. State explicitly what you inherited and what you built or modified.

What if my experiments fail or my hypothesis is wrong?

This is fine and even expected sometimes in research. What matters is how you analyze and discuss the failure. Why did it fail? What does this tell us? What would you try next?

14. Summary of Deliverables Checklist

Before submitting, verify that you have:

- ☐ Written report (PDF), compiled from the NeurIPS L^AT_EX template, within page limits.
- ☐ Code (repository link or .zip) with README and dependency file.
- ☐ One presentation slide (PDF or PPTX).
- ☐ Contribution statement (if working in a pair).
- ☐ AI tools disclosure statement (if applicable).
- ☐ All references properly cited.